

Solution Requirements

Date: 15 March 2026

Team ID: (xxxxxxx)

Project Name: Human Development Index (HDI) Prediction Using Machine Learning

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	The system allows authorized users or administrators to securely access the application. General users can directly use the prediction interface without registration.	High
2	Authorization Levels	The administrator can update the dataset, retrain the model, and manage the application. End users can only enter input values and view prediction results.	High
3	External Interfaces	The application interacts with the HDI dataset (CSV file), Flask web interface, Scikit-learn machine learning model, and Joblib model file (model.pkl).	High
4	Transaction Processing	The system accepts Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI Per Capita as input, processes the data, and predicts the HDI category.	High
5	Reporting	The application displays the predicted HDI category and may generate charts or summary reports for analysis.	Medium
6	Business Rules	The system validates all input values before prediction. Predictions are generated only when all required inputs are provided. The model classifies the country into one of four categories: Very High, High, Medium, or Low HDI.	High

S.No	Requirement Category	Requirement Description	Priority
7	Compliance to Laws or Regulations	The system uses publicly available HDI datasets and follows basic data privacy and ethical AI principles.	Medium
8	Other	The application provides a simple, user-friendly interface with clear instructions, accurate predictions, and error messages for invalid input.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should process user input and generate predictions quickly.	Prediction displayed within 2 seconds .
2	Scalability	The application should support future enhancements such as additional datasets, more users, and cloud deployment.	Supports multiple users and future expansion.
3	Security & Data Privacy	User input data should be processed securely. The trained model and dataset should be protected from unauthorized modification.	Secure handling of input data and model files.
4	Reliability & Availability	The application should provide consistent predictions with minimal downtime.	99% system availability during normal operation.
5	Usability & Accessibility	The interface should be easy to use for students, researchers, and policymakers with clear labels and navigation.	Responsive interface with simple input forms.
6	Maintainability & Portability	The application should be easy to update, debug, and deploy on Windows, Linux, or macOS platforms.	Modular code and cross-platform compatibility.

Team Members and Responsibilities :-

Team Member	Role	Responsibility
G Naveen	Team Lead	Project planning, GitHub management, model integration
Chandana Busireddy	Member	Dataset collection, preprocessing, and documentation
Gandham Joshna	Member	Data visualization and exploratory data analysis
Nandineni Gowthami	Member	Flask web application development and testing
Shaik Talha	Member	Machine learning model training, documentation, GitHub updates, and presentation

Expected Outcome :-

The **Human Development Index (HDI) Prediction System** enables users to predict the HDI category of a country using four key indicators: **Life Expectancy**, **Mean Years of Schooling**, **Expected Years of Schooling**, and **GNI Per Capita**. The system provides quick, accurate, and user-friendly predictions, helping governments, researchers, students, and policymakers analyze development levels and support informed decision-making.